How Small is Small?

Non-linearities in Heterogeneous Agent Models

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Overview

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  - Capture salient features of household spending behavior
  - Match the level and distribution of MPCs in the data

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- BUT: How small is small?
- Our answer: for fiscal transfers, "small" must be *very* small.

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- Local methods are unreliable when aggregate responses are driven by failures of Ricardian equivalence
  - Shocks where they *are* accurate are those where HA $\approx$ RA.

Roadmap

1. Partial equilibrium model and calibration
2. Fiscal stimulus in general equilibrium
3. Other shocks

Environment: Household Problem

$$\max_{\{c_{i,t}, a_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_i^t u(c_{i,t})$$

subject to

$$c_{i,t} + a_{i,t} \leq (1+r)a_{i,t-1} + (1-\tau)y_{i,t} + T_t \quad \text{and} \quad a_{i,t} \geq 0.$$

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Intertemporal Euler equation:

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with equality if $a_{i,t} > 0$.

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Aggregate consumption function:

$$\mathcal{C}_t(\{T_s\}) = \int c_t(a,y,\beta)d\Gamma_t(a,y,\beta)$$

Fiscal Transfers

Starting from steady state, a one-time lump-sum transfer at $t = 0$:

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Non-linear solution

- Solve individual household problem exactly and aggregate

Linear solution

- Taylor expansion of aggregate consumption function around steady state

$$\mathcal{C}_t^{\text{lin}}(\Delta) = \mathcal{C}_t(0) + \left.\frac{\partial \mathcal{C}_t}{\partial \Delta}\right|_{\Delta=0} \cdot \Delta$$

Calibration

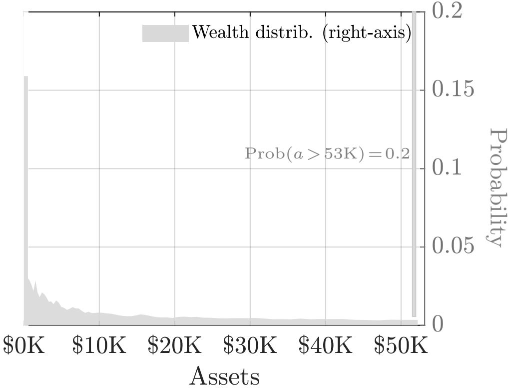
- A period = a quarter. Normalize average quarterly income to 1.
- Earnings process, $y_{i,t}$, subject to temporary and persistent shocks
  - Calibrated as in Kaplan-Violante (2022) to match distribution of households' earnings
- Interest rate 2% annual.
- Tax rate (τ) = 27.4%
- Steady-state transfers = 15% of GDP (≈ \$25K/yr)

Calibrated to match moments

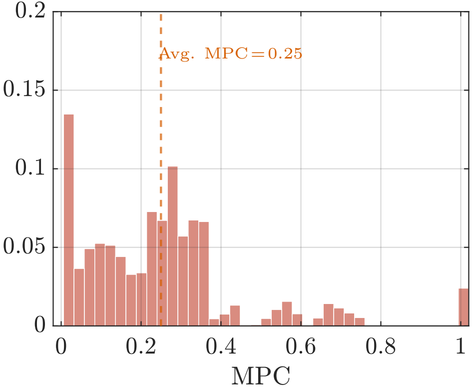
$$\left.\begin{array}{l}\text{Mean wealth / annual GDP} = 100\% \ (\approx \$166\text{K}) \\ \text{Avg. quarterly MPC (out of \$1,000)} = 25\%\end{array}\right\} \Rightarrow \bar{\beta} = 0.972,\ \sigma_\beta = 0.010$$

Steady-state wealth distribution and MPCs

(a) Wealth distribution



(b) MPC distribution



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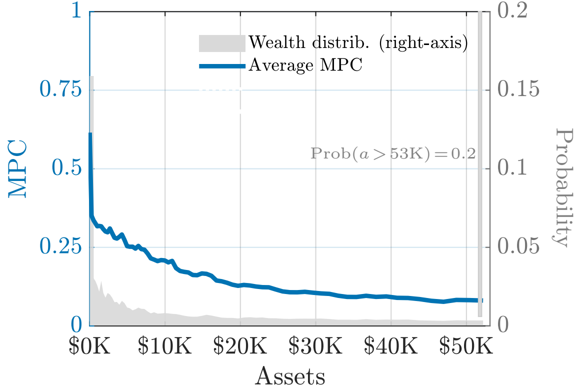

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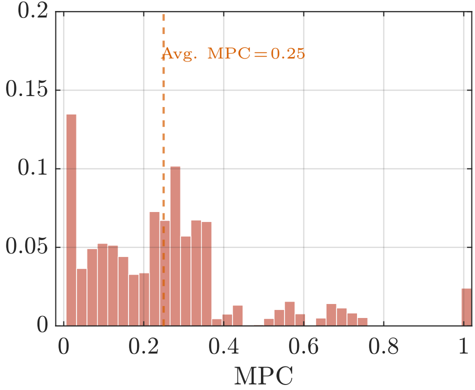

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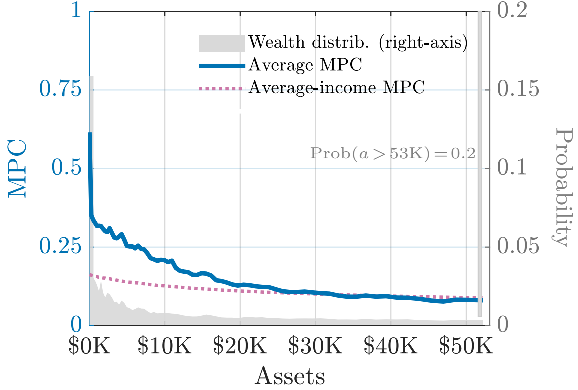

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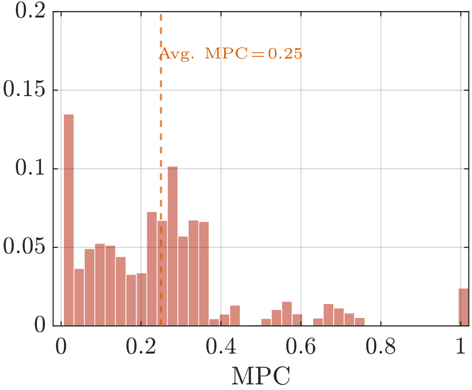

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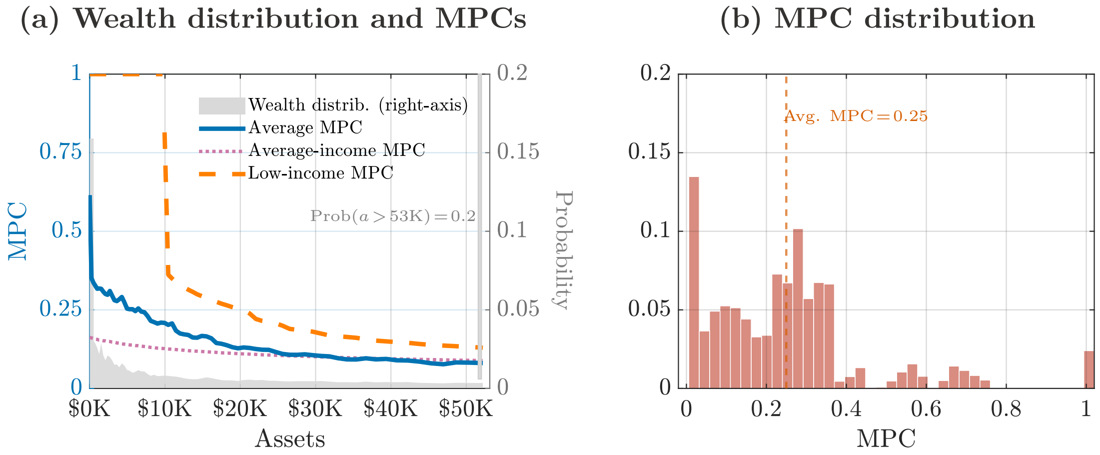
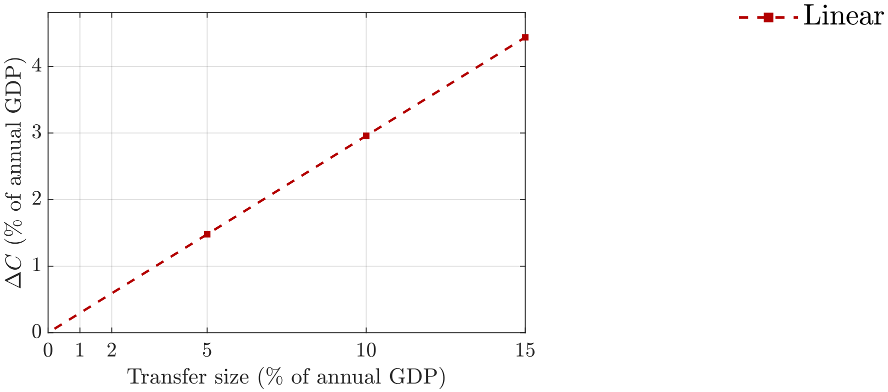


Aggregate consumption response



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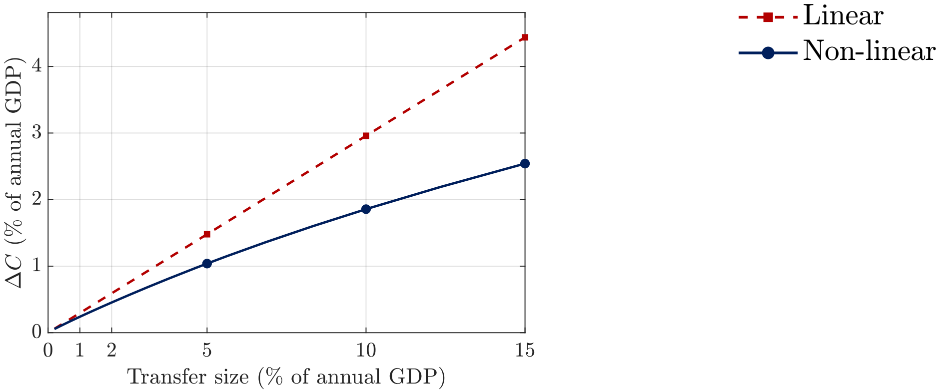

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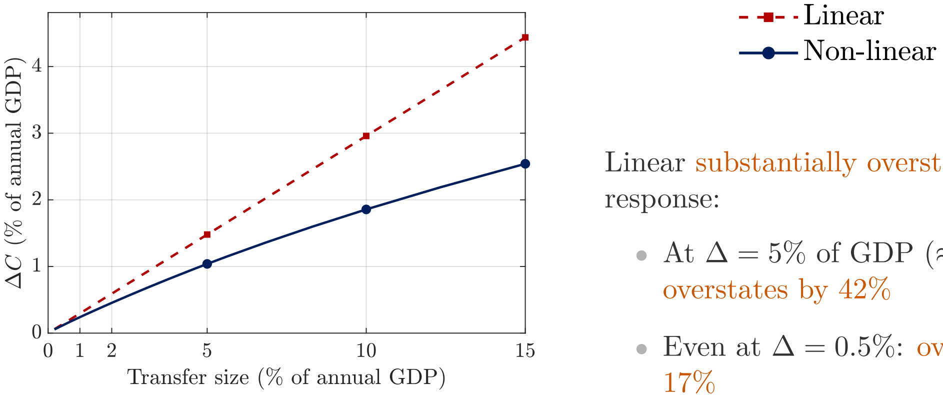

Linear substantially overstates response:

- At $\Delta = 5\%$ of GDP ($\approx$ \$8,300): overstates by 42%
- Even at $\Delta = 0.5\%$: overstates by 17%

Concavity and the Kink

- MPC is decreasing in wealth and drops discontinuously at the kink

$$m(x,y) = \begin{cases} 1, & x < \bar{x}(y), \\ m^+ & x > \bar{x}(y). \end{cases}$$

$$\frac{\beta\,\mathbb{E}\big[\,V_{aa}\big(a'(x,y),y'\big)\mid y\,\big]}{u''(c)+\beta\,\mathbb{E}\big[\,V_{aa}\big(a'(x,y),y'\big)\mid y\,\big]}$$

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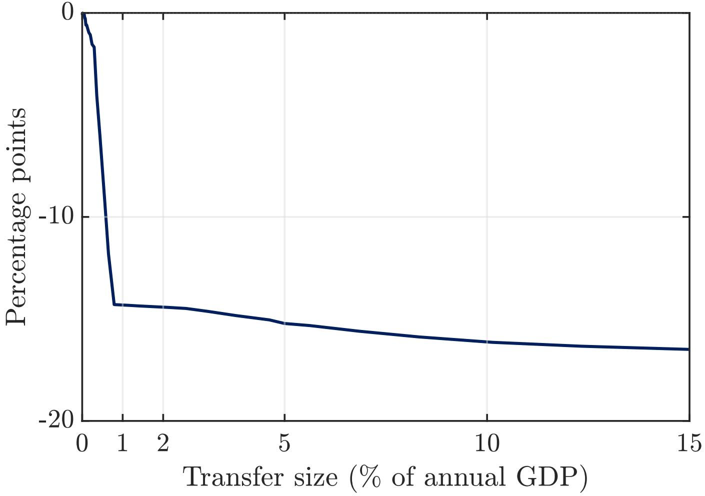
- Aggregate MPC $\downarrow$ as $\Delta \uparrow \Rightarrow$ concave aggregate consumption response

Change in share of constrained households



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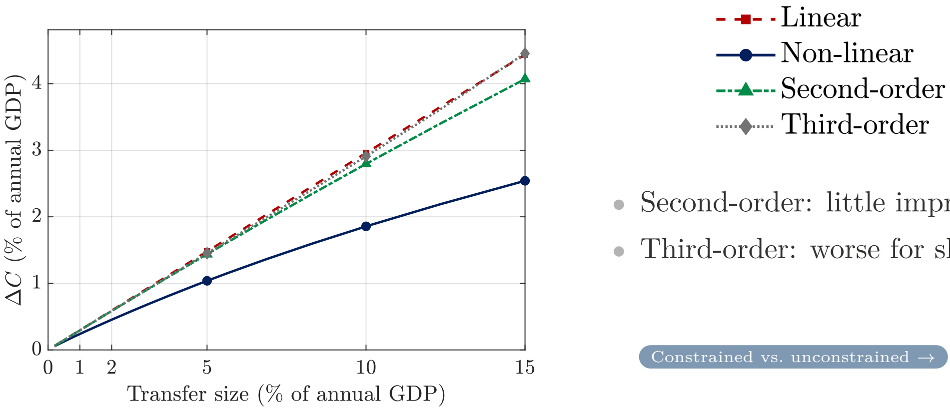
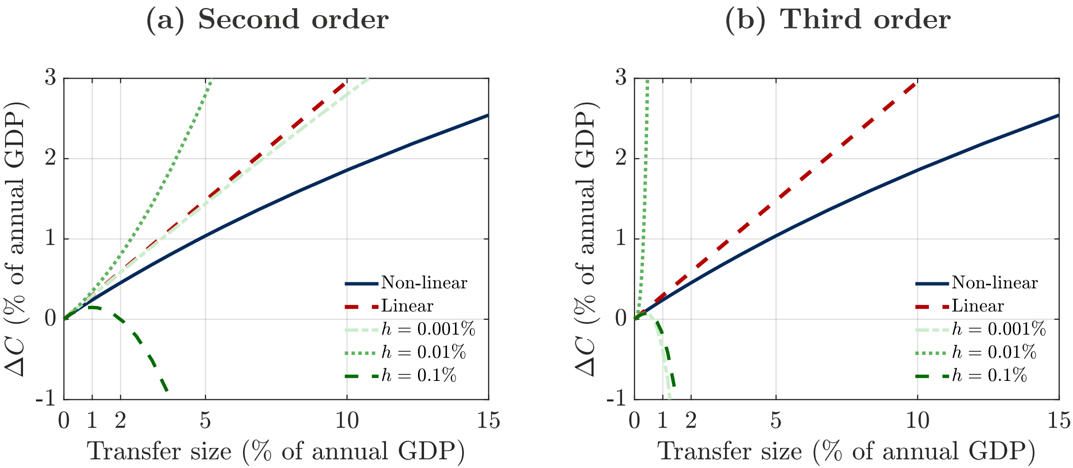


- Second-order: little improvement
- Third-order: worse for shocks $> 5\%$

Constrained vs. unconstrained →

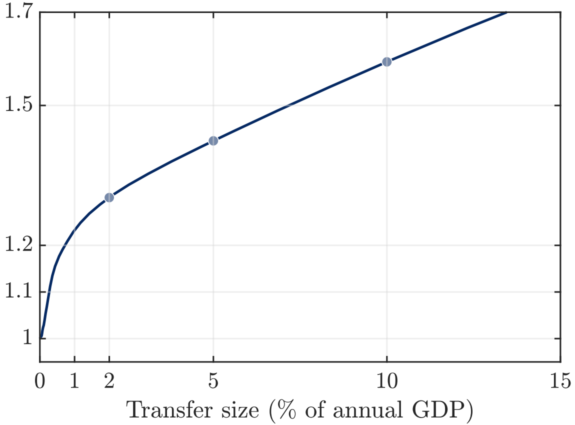
Higher-order with finite differences



Extremely sensitive to step size and poorly behaved even for small shocks.

Robustness of Partial Equilibrium Results

Linear/Non-linear Consumption Response



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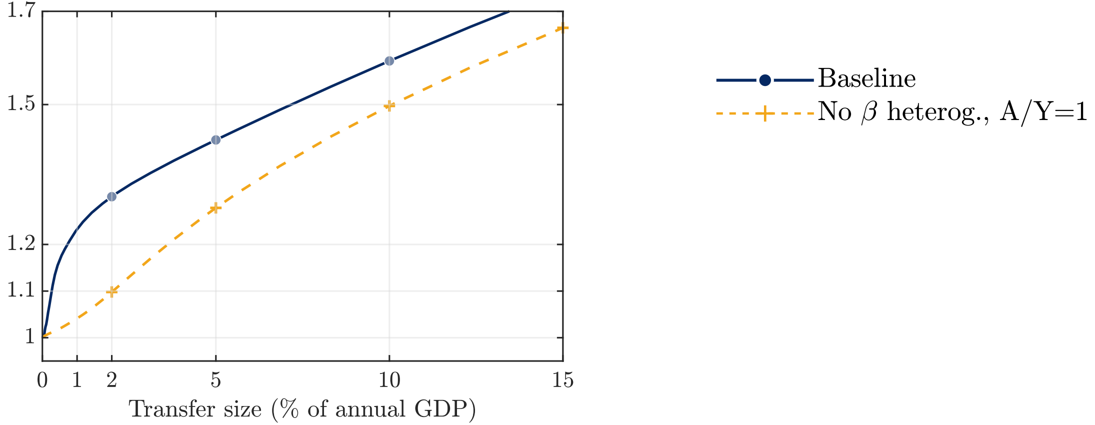

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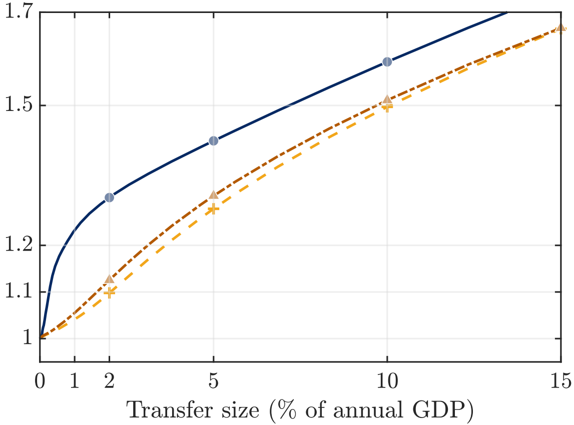

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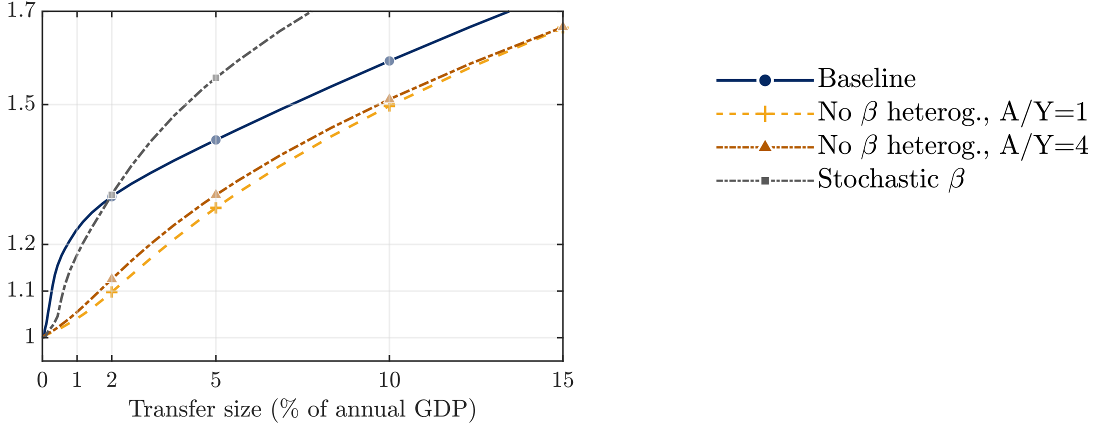

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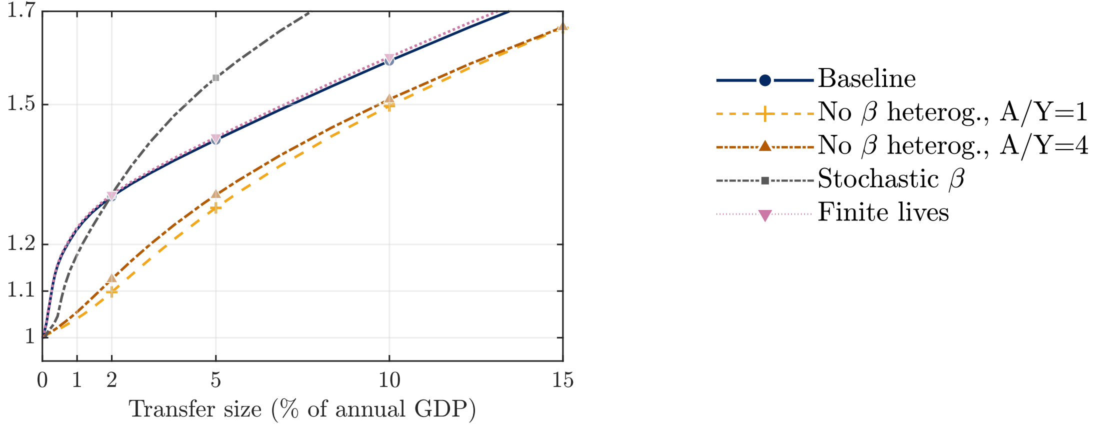

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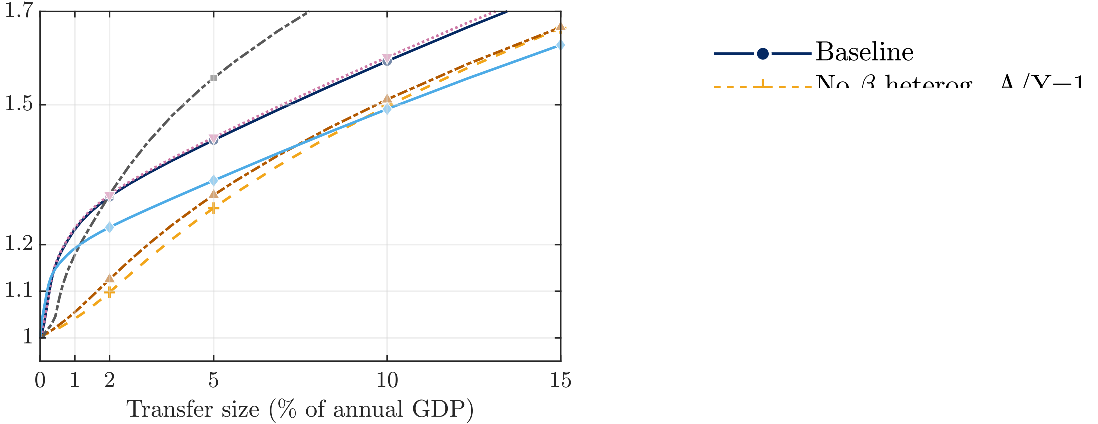

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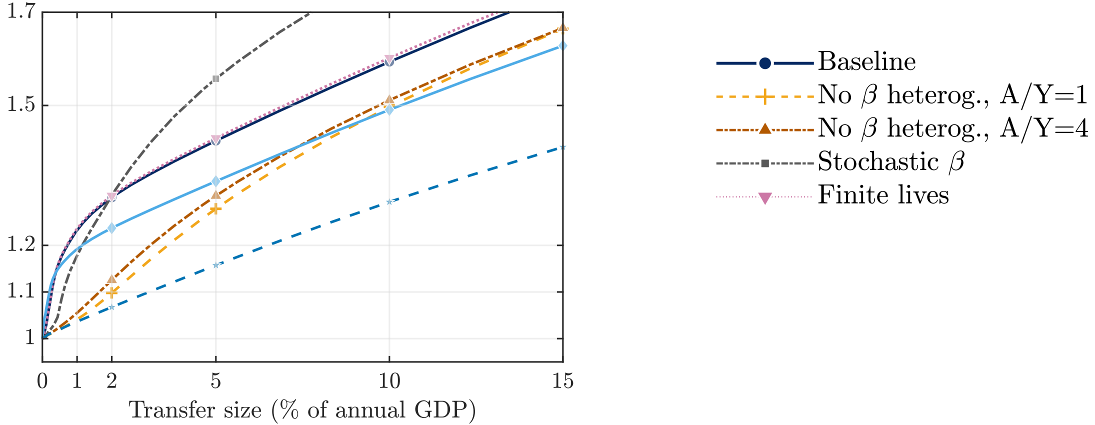

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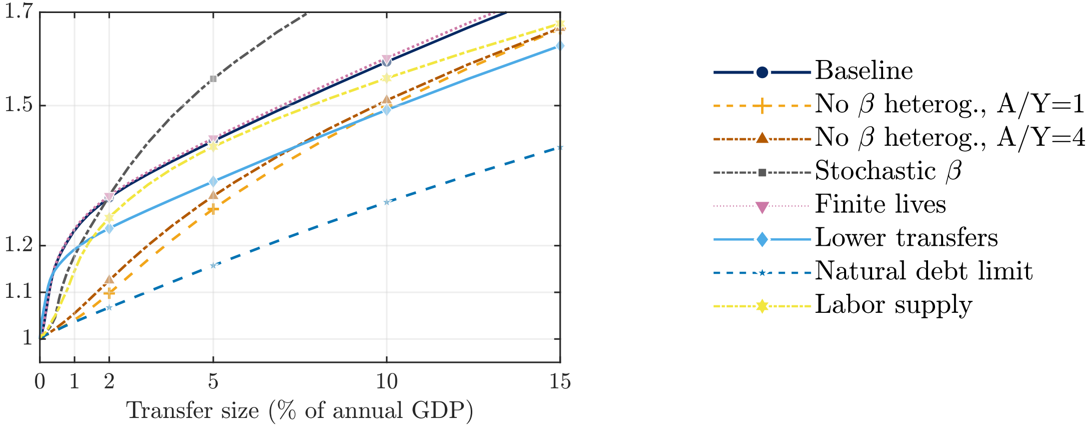

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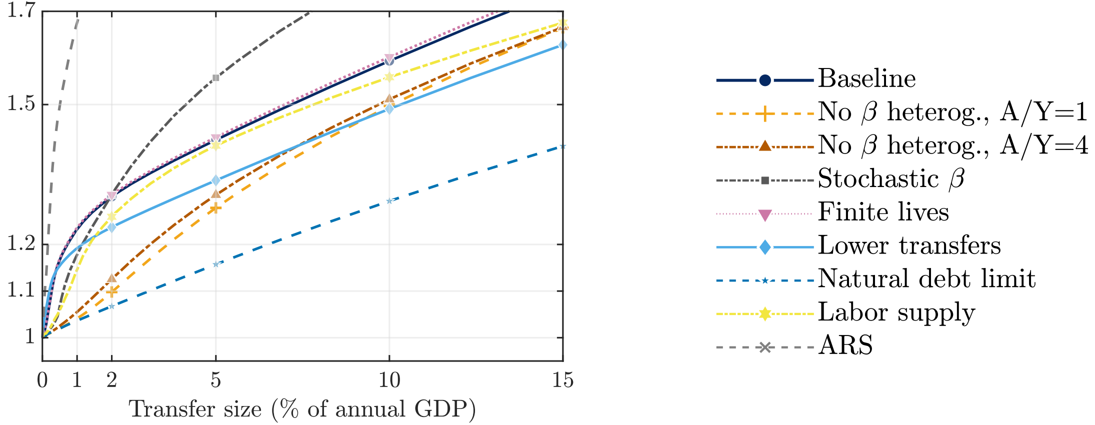

Roadmap

1. Partial equilibrium model and calibration
2. Fiscal stimulus in general equilibrium
3. Other shocks

General Equilibrium Model

PE + production, labor supply, nominal rigidities, fiscal/monetary rules.

General Equilibrium Model

- **Households:**
  - Utility flow, $u(c_{it}) - v(n_{it})$ with $v' > 0, v'' > 0$
  - Income $(1-\tau)\left(\frac{W_t}{P_t} e_{it} n_{it} + d_{i,t}\right)$

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  - Labor union with sticky wages (linearized)

$$\pi_t = \kappa \left(v'(N_t) - \frac{1-\tau}{C_t} \right) + \bar{\beta}\, \pi_{t+1}, \qquad \text{(Phillips Curve)}$$

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- **Firms and Technology:**

$$\left.\begin{array}{l} \text{Production: } Y_t = N_t \\ \text{Flexible prices} \end{array}\right\} \Rightarrow W_t = P_t, \quad d_{i,t} = 0$$

General Equilibrium Model

Government Policies

General Equilibrium Model

Government Policies

- Government budget:

$$B_t + \tau Y_t = G + T_t + (1 + r_{t-1})B_{t-1}$$

- Fiscal rule ($t > 0$)

$$T_t = T - \phi_B \bar{r}(B_{t-1} - \bar{B})$$

- Monetary policy:

$$i_t = \bar{i} + \phi_\pi(\pi_t - \bar{\pi})$$

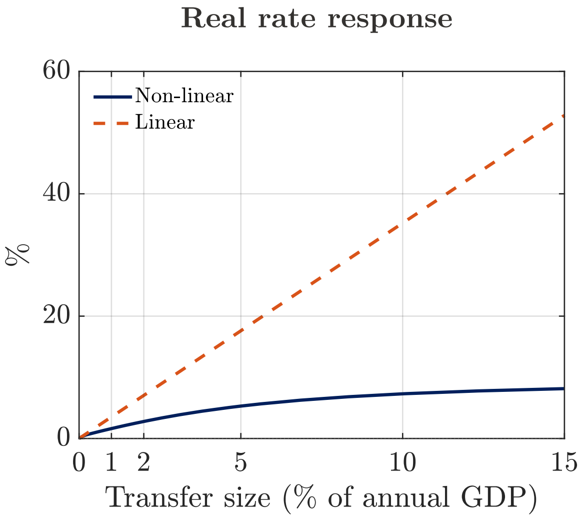
Calibration

Parameter	Value	Description
$\bar{B}$	4	Steady-state government debt=100% GDP
$\bar{r}$	0.02	Steady-state annual real interest rate
$\bar{\pi}$	0	Steady-state inflation
κ	0.01	Phillips curve slope
ν	1	Frisch elasticity of labor supply
ϕ_π	1.5	Taylor coefficient
ϕ_B	8	Fiscal rule
G	0.104	Government spending
T	0.15	Government Transfers

Computation:

- Linear: sequence space Jacobian (ABRS)
- Non-linear: quasi-Newton algorithm

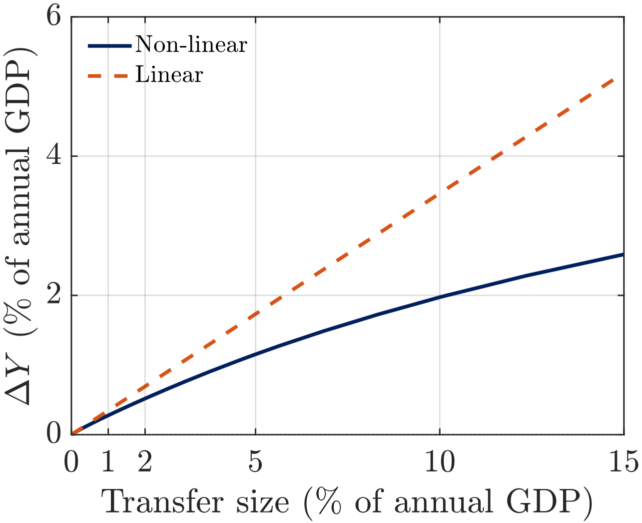
Flexible-Price Allocation: Real Interest Rate



- No output effects
- Linear methods overstate increase in demand and rise in r

With household labor choice →

HANK: Output Response

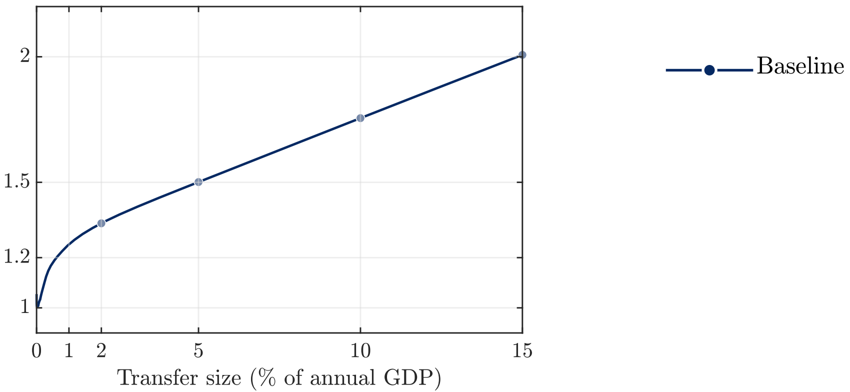


- At $\Delta = 2\%$ GDP: linear overstates by 34%
- At $\Delta = 5\%$ GDP: linear overstates by 50%

Impulse responses →

Robustness in HANK

Linear/Non-linear Output Response



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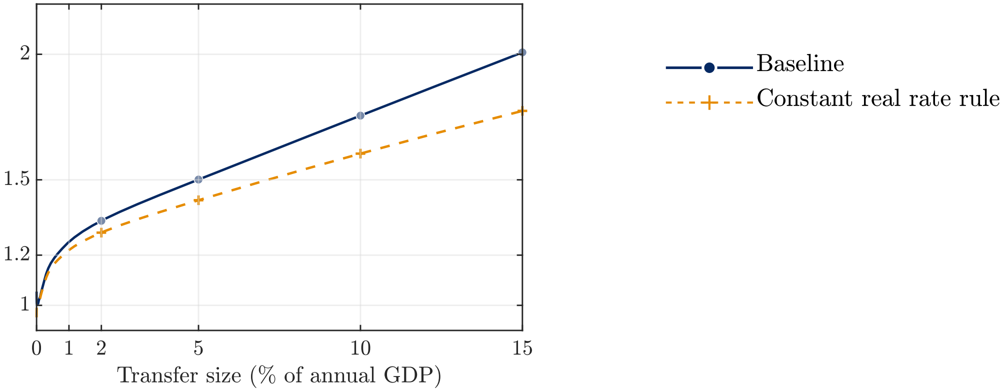

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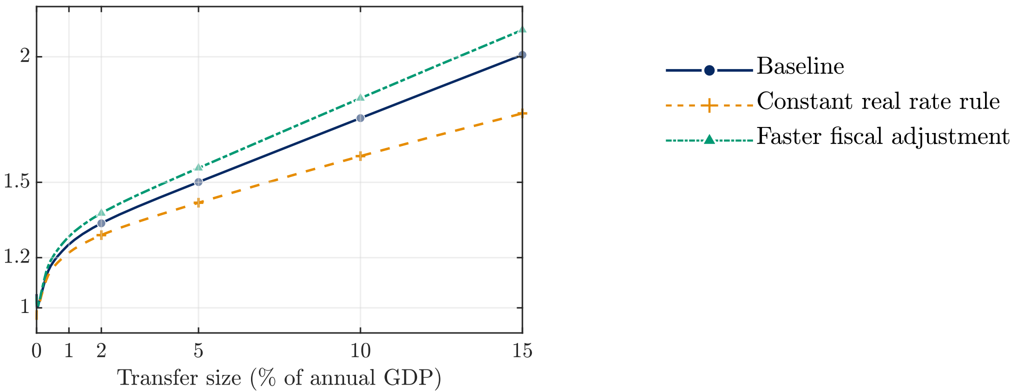

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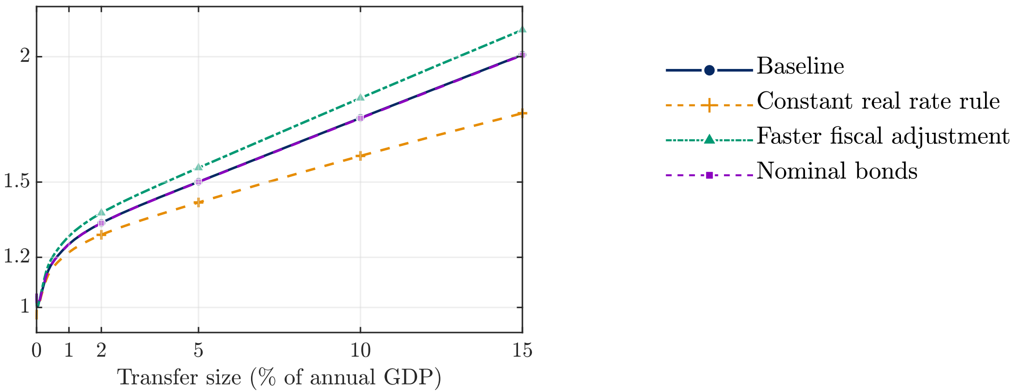

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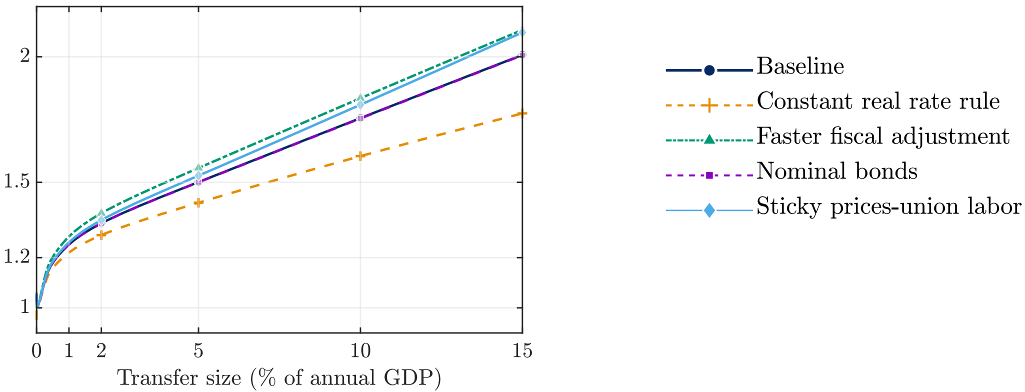

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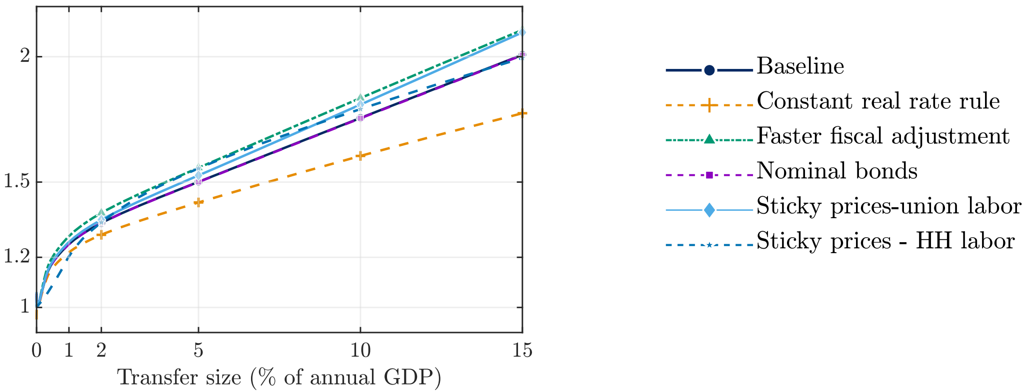

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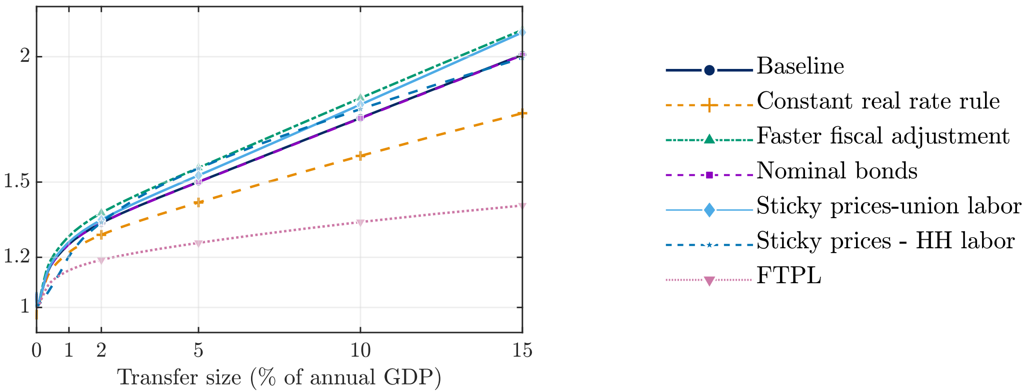
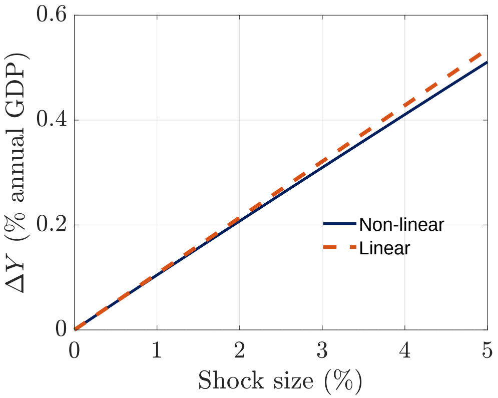
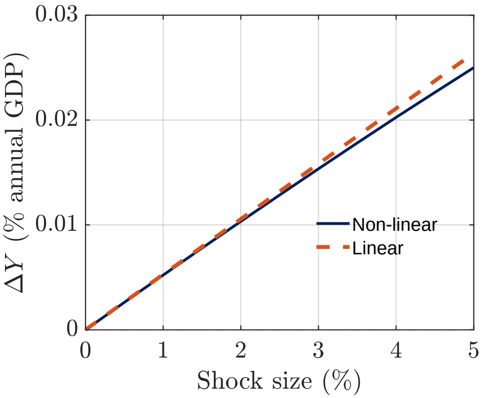


Non-linearities for monetary policy and TFP

(a) Monetary (rate cut)



(b) TFP



Conclusions

- HA models are valuable for analyzing fiscal and monetary policy because they capture realistic household behavior

- For monetary policy, linear approximation is accurate. But HA $\approx$ RA

- For fiscal stimulus: need non-linear methods
  - Works through departures from Ricardian equivalence, where MPCs are strongly non-linear in wealth:
  - First-order methods (SSJ) treat constrained households as permanently $MPC = 1$, overstating their true response. Higher-order local methods do not fix this

For local methods to be reliable for fiscal stimulus, "small" must be *very* small.

Appendix

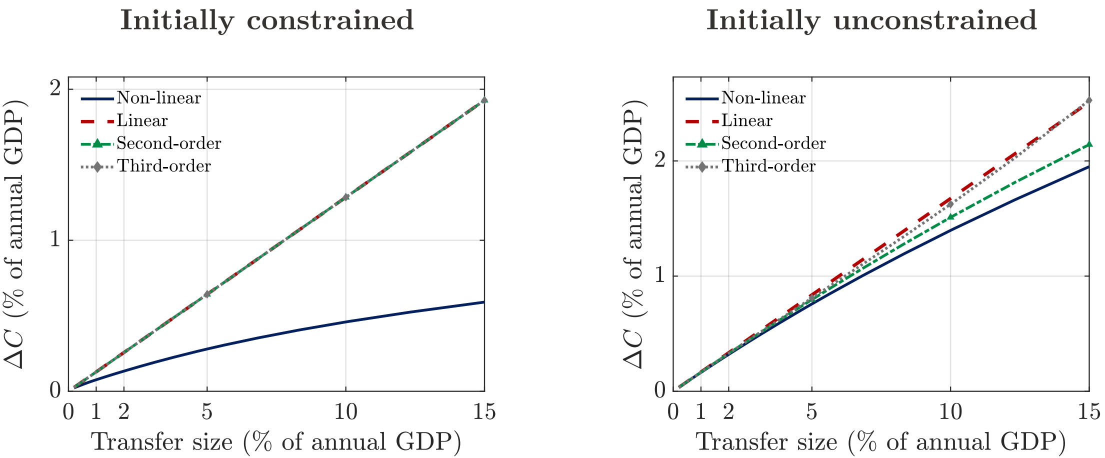
Additional figures

Implied parameters

Parameter	Value
Preferences & assets	
r (annual)	2%
$\bar{\beta}$ (mean disc. factor)	0.972
σ_β (dispersion)	0.010
Stochastic income process *(Kaplan–Violante 2022)*	
ϕ_z (AR persistence)	0.988
σ_η^2 (perm. var.)	0.044
σ_ε^2 (trans. var.)	0.638
λ (shock arrival)	1/4 per qtr

◂ Back to calibration

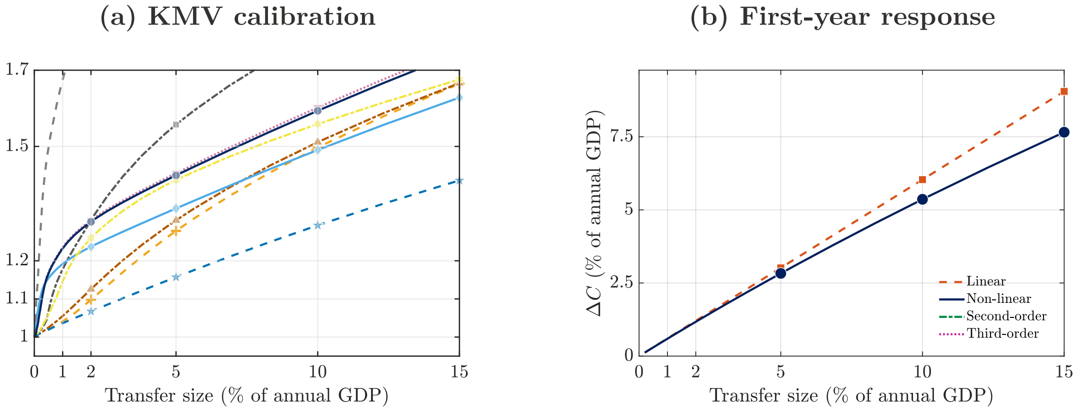
Constrained vs. Unconstrained



Note: Consumption response within each group, as a % of annual GDP per capita.

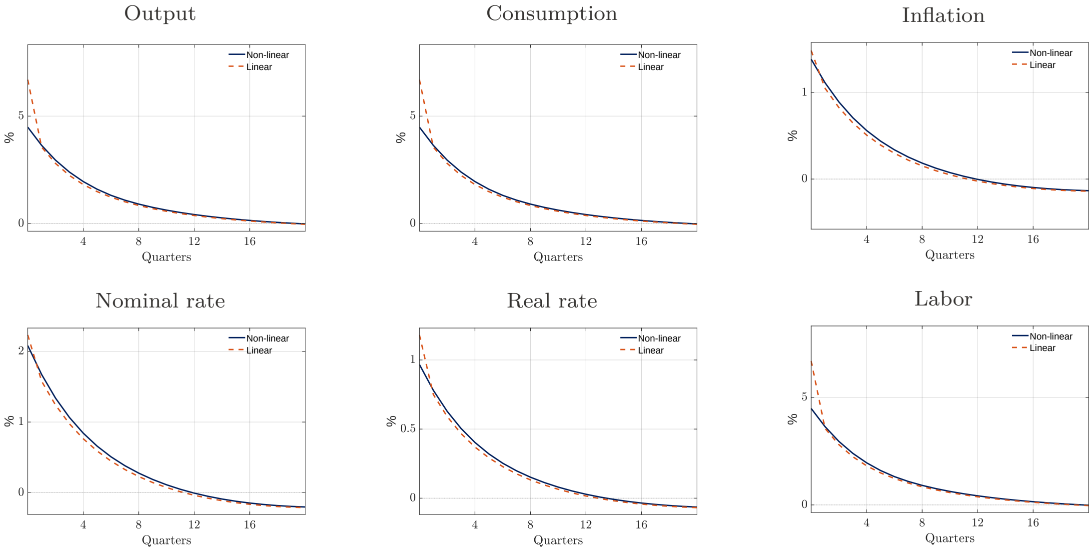
Back

PE: KMV calibration and first-year response

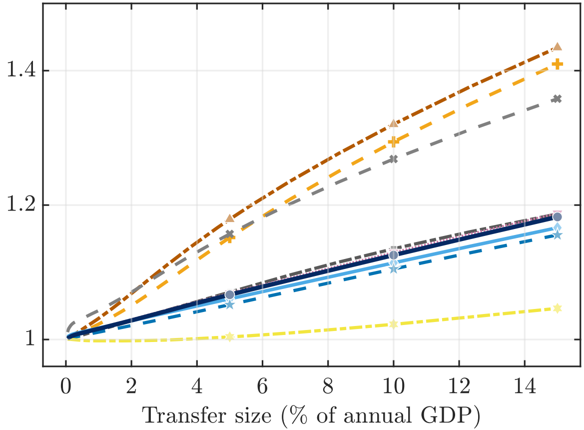


Note: Concavity is robust to the (Kaplan, Moll & Violante 2018) calibration and remains over the first year (panel b), though smaller than on impact.

GE baseline: impulse responses

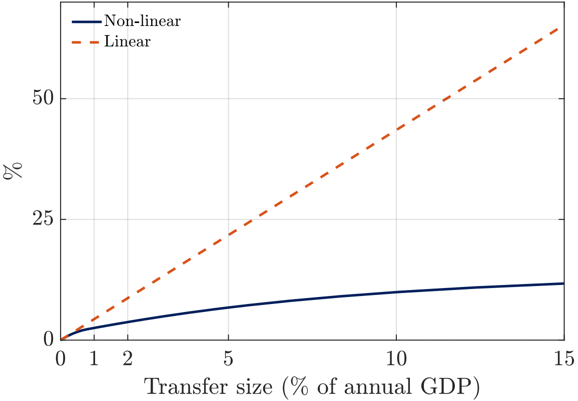


GE: first-year linear-to-non-linear ratio



Note: Over the first year the linear method still overstates the output response, though by less than on impact.

GE: flexible prices with household labor choice



- Households choose labor freely (no union)
- Small output effects via labor-supply incentives
- Real-rate response remains concave in Δ — same mechanism as the union case

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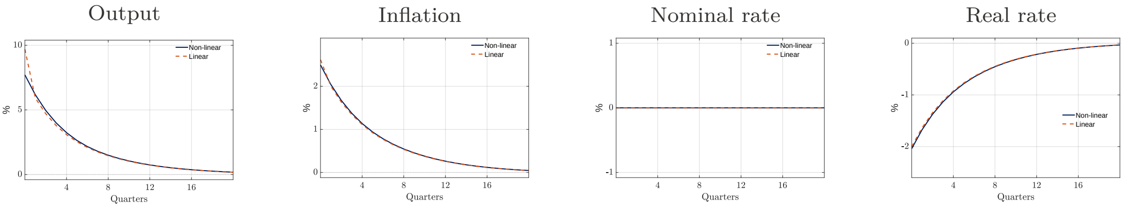
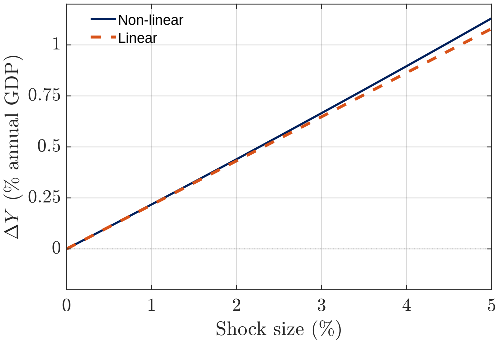
GE: unfunded stimulus (FTPL) impulse responses



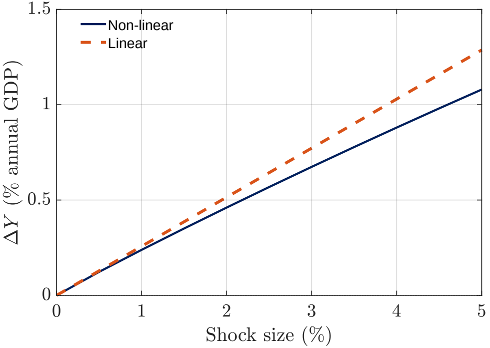
Note: Under the FTPL (active fiscal, passive money), the output effect does not hinge on a failure of Ricardian equivalence, so non-linearity is much milder.

Other shocks: discount factor and labor tax

(a) Discount-factor shock



(b) Labor-tax cut



Note: Both generate milder non-linearity than lump-sum transfers; a τ cut shifts fewer resources to low-income, constrained households.